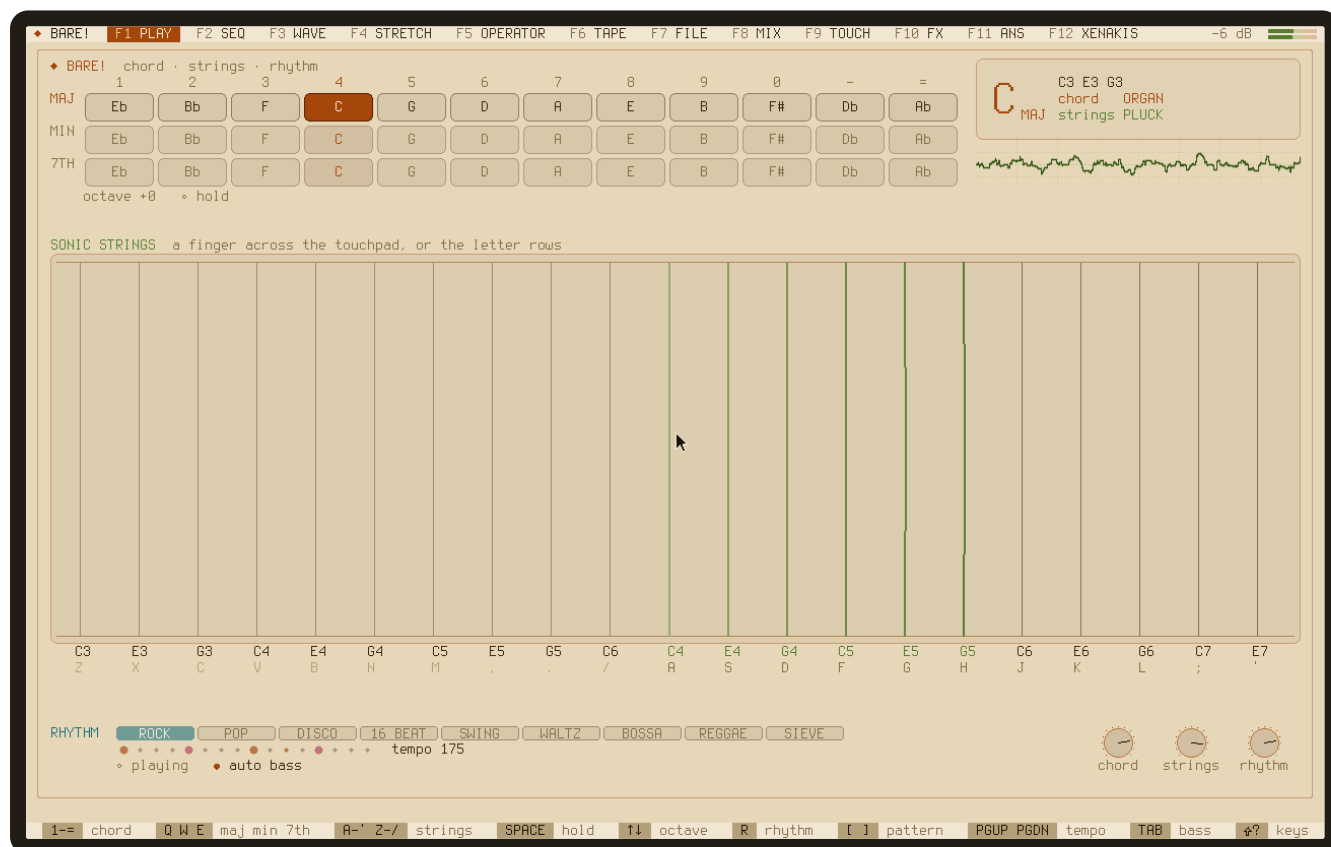


BARE!

a synthesizer with no operating system

F1 PLAY	F2 SEQ	F3 WAVE	F4 STRETCH	F5 OPERATOR	F6 TAPE
F7 FILE	F8 MIX	F9 TOUCH	F10 FX	F11 ANS	F12 XENAKIS



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BARE !

WHAT IT IS

A synthesizer that is the whole computer. It starts from a USB stick straight into the instrument, with no operating system underneath, on PCs from the late 1990s to now.

It has an omnichord with a rhythm section, a tracker, a Fairlight-style wave page with a sampler, 4-operator FM, a freezer that stretches what you just played, an 8-track tape, a mixer with effects, a crackle box played with your fingers, effects to play live, Murzin's ANS, four of Xenakis's ways of making music (UPIC, GENDY, sieves and clouds), instruments you write as text files, MIDI and Ableton Link. And Doom.

2.5 is a beta. It has been played on a ThinkPad X250 and an ASUS VivoBook (Intel 12th gen), and it boots on 89 emulated PCs. Not tried on real hardware yet: AC'97, the Sound Blaster 16, SATA and NVMe disks, the installer and Ethernet. It needs a Pentium Pro or later.

1. Download `bare-2.5.img.xz` from the [2.5 release](#).
2. Write it to a USB stick of 256 MB or more with balenaEtcher or Raspberry Pi Imager, or on Linux with `xz -dc bare-2.5.img.xz | sudo dd of=/dev/sdX bs=4M conv=fsync`. Everything on the stick is replaced.
3. Turn Secure Boot off, plug the stick in and start the computer from it: the boot menu is usually `[F12]`, `[F9]`, `[F8]` or `[Esc]` at power-on.
4. When the splash plays, press any key. Hold `[4]` and run a finger along the keys from `[A]` to `[']`: a C major chord, strummed.

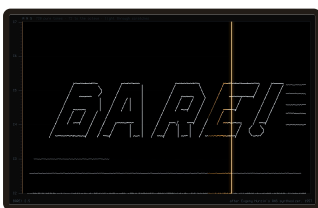
It runs from memory, so once it has started the stick can come out; projects are saved on the stick. The installer (on the FILE page) erases the disk you choose.

Free software: GPL-3.0-or-later ([LICENSE](#)); [NOTICE.md](#) lists what it includes from others.

STARTING UP

THE BOOT AND ITS SPLASH

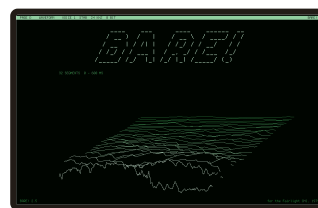
While the machine starts, the name and the boot log are on screen (a machine that stops during boot shows where). Then comes a short splash, a different one each time the stick can remember the last: ANS (the name scratched into a glass plate and played by light, after Murzin's photoelectronic synthesizer), GENDY (Xenakis's dynamic stochastic synthesis: waveforms that random-walk, then settle into a chord), CMI (for the Fairlight: a green terminal, 8-bit orchestra stabs made here, a 3D waveform) and METASTASEIS (Xenakis's 46 string glissandi drawn on graph paper). Any key, click or MIDI note ends it; the sound follows the saved volume and mute.



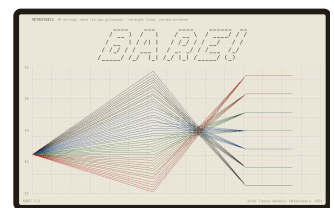
ANS



GENDY



CMI



METASTASEIS

KEYS

THE F KEYS, SHIFT AND THE REST

The F keys switch pages and do nothing else: **F1** play · **F2** tracker · **F3** waves and sampler · **F4** stretch · **F5** operator (FM) · **F6** tape · **F7** projects · **F8** mixer · **F9** touch · **F10** effects · **F11** ANS · **F12** Xenakis. On keyboards without an F row, **Ctrl+1...9**, **Ctrl+0**, **Ctrl+[-]** and **Ctrl+[=]** do the same. (On ThinkPads **Fn+Esc** locks the F keys.)

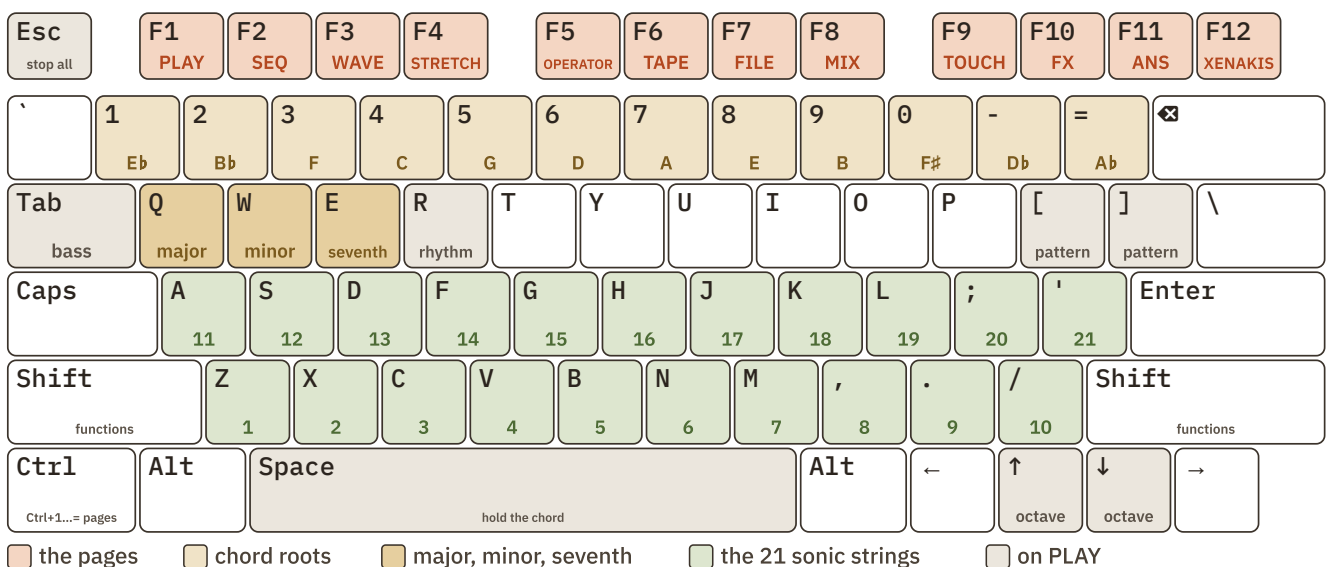
Shift is the function layer, on every page: **Shift+[↑↓]** chord sound · **Shift+[←→]** strum sound · **Shift+[B]** 1-bit · **Shift+[E]** echo · **Shift+[F]** freeze (stretch) · **Shift+[T]** thermal · **Shift+[-]** / **Shift+[=]** volume · **Shift+[M]** mute · **Shift+[H]** colours · **Shift+[?]** all of these on screen. **Esc** stops everything and releases all notes.

Colours: **Shift+[H]** goes through ten colour schemes: NIGHT (the first one), PHOSPHOR (green on black), AMBER, PAPER (ink on paper), BLUEPRINT, LCD (a handheld's olive greens), STAGE (black, white and yellow, for projectors and sunlight), CREAM (an Omnicord's cream and brown), ACID and SUNSET. The stick keeps the choice, with volume and mute. The SCAN wave reads the screen, so it sounds a little different in each.

Ctrl+[Z] undoes and **Ctrl+[Y]** redoes (or **Shift+[Z]**, **Shift+[Y]**): tracker edits and the order list, wave drawing and harmonics, FM changes, sample edits, tape takes, erases, pastes and imports. A pen stroke or a run of the same key is one step. How far back it goes depends on the memory: 1.5 MB of history on a 32 MB PC, up to 64 MB; a tape take too long for it can't be undone, and the title bar says so.

Volume: the laptop's volume and mute keys work too. It starts at -6 dB (the title bar shows it): built-in speakers distort when the sound chip is driven at full level. Volume, mute and 1-bit are kept on the stick with the next project you save or load.

The key strip at the bottom of each page shows its main keys. A page's own keys never also play a chord or a strum; the keys it doesn't use still do.



The keyboard on PLAY, the first page: the F keys go to the pages everywhere.

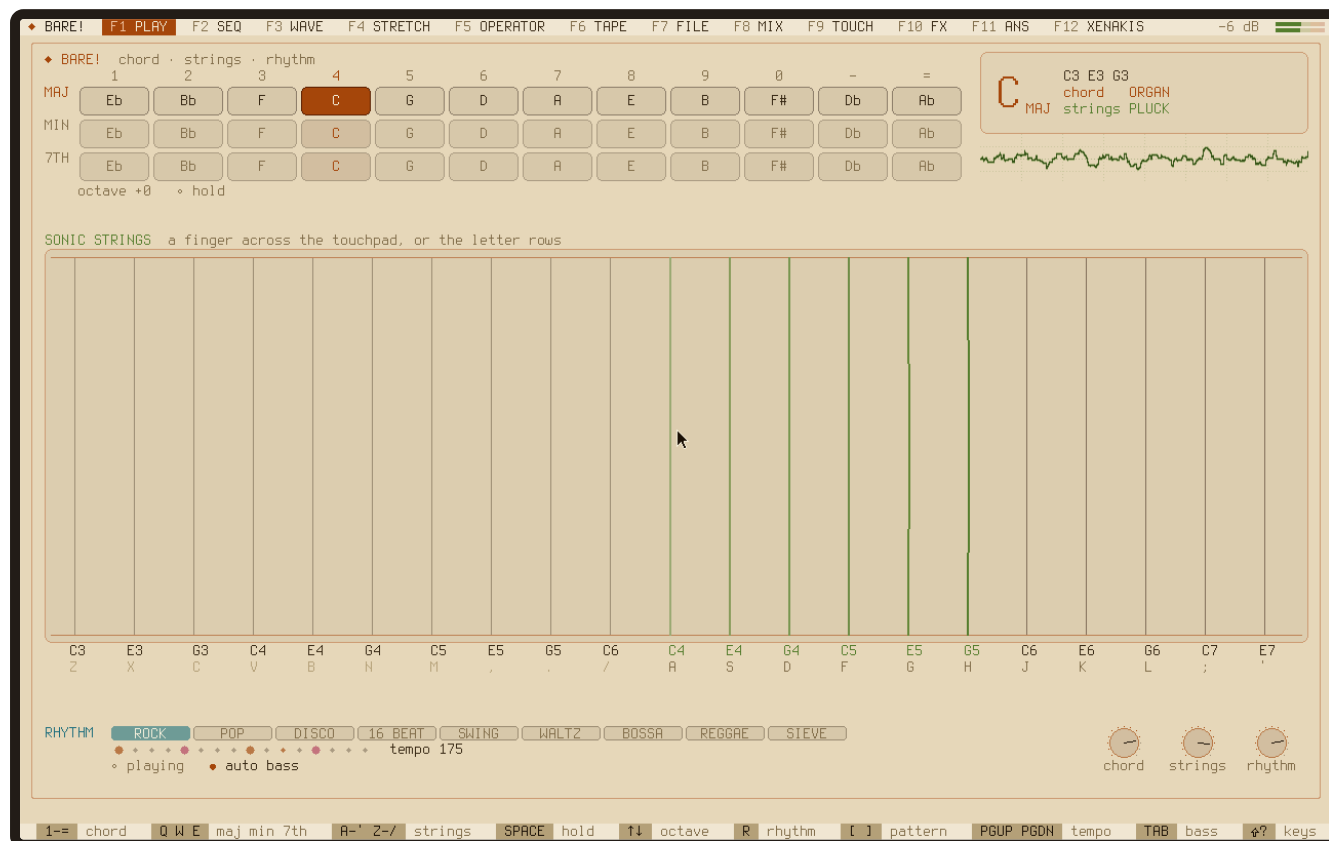
ECHO

Echo (**Shift+[E]**) is a stereo ping-pong: repeats alternate left and right, a dotted eighth apart at the sequencer's tempo, and get darker as they go. The output ends in a limiter, so loud chords and strums get quieter instead of clipping. The meter at the right of the title bar is the output, left and right.

THERMAL MODE

Thermal mode (**Shift+[T]**)—the CPU's temperature sensor drives filter cutoff and detune, and while it's on the idle loop spins instead of halting, so the machine warms up as you play. On ThinkPads the EC also gives fan RPM and the power/lid LEDs blink on the sequencer beat.

An omnichord, drawn in lines and laid out like the keyboard: chord buttons on top, the sonic strings below, a display with the chord, its notes, the two sounds and the scope, and the rhythm section.



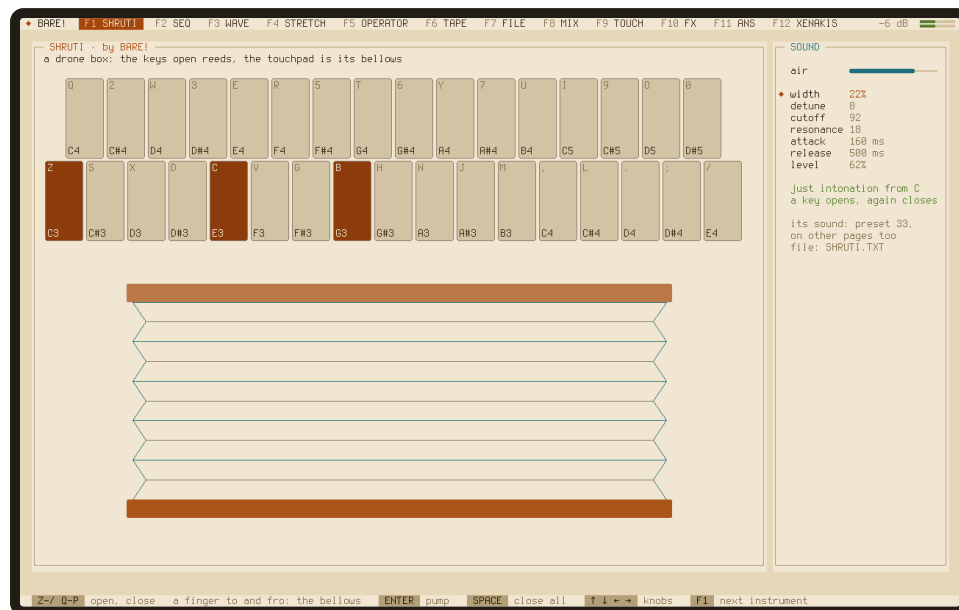
F1 · PLAY

KEYS

[1] ... [=]	chord root (Eb Bb F C G D A E B F# Db Ab—circle of fifths, like the Omnichord)—plays while held
[Q] [W] [E]	major / minor / seventh
[Z] ... [/] then [A] ... [']	the 21 sonic strings, low to high (the chord's tones over three octaves)
touchpad	fingers across it strum the strings, left low to right high, several at once (I2C precision touchpads: up to five; Synaptics PS/2 pads: two)
[Space]	hold the chord after releasing the button
[↑] [↓]	strings octave
[Shift] + [↑↓] / [Shift] + [↔]	chord sound / strings sound (ORGAN, PLUCK, SAW BASS, FAT SAW, SQ LEAD, CHIP, BELL, GRIND, KICK, SNARE, HAT, the wave and FM sounds, and the samples that hold something)
[R]	rhythm start / stop
[L] [J]	rhythm pattern: ROCK, POP, DISCO, 16 BEAT, SWING, WALTZ, BOSSA, REGGAE, and SIEVE (the XENAKIS page's sieves)
[PgUp] [PgDn]	tempo (shared with the sequencer, which the rhythm falls in step with)
[Tab]	auto bass: a bass that follows the chord you hold, in the pattern's rhythm

With the pointer: press a chord button to play it; drag across the strings to strum them; click a pattern or the switches; drag a knob up or down (chord, strings and rhythm levels). The rhythm, its settings and the levels are saved with the project.

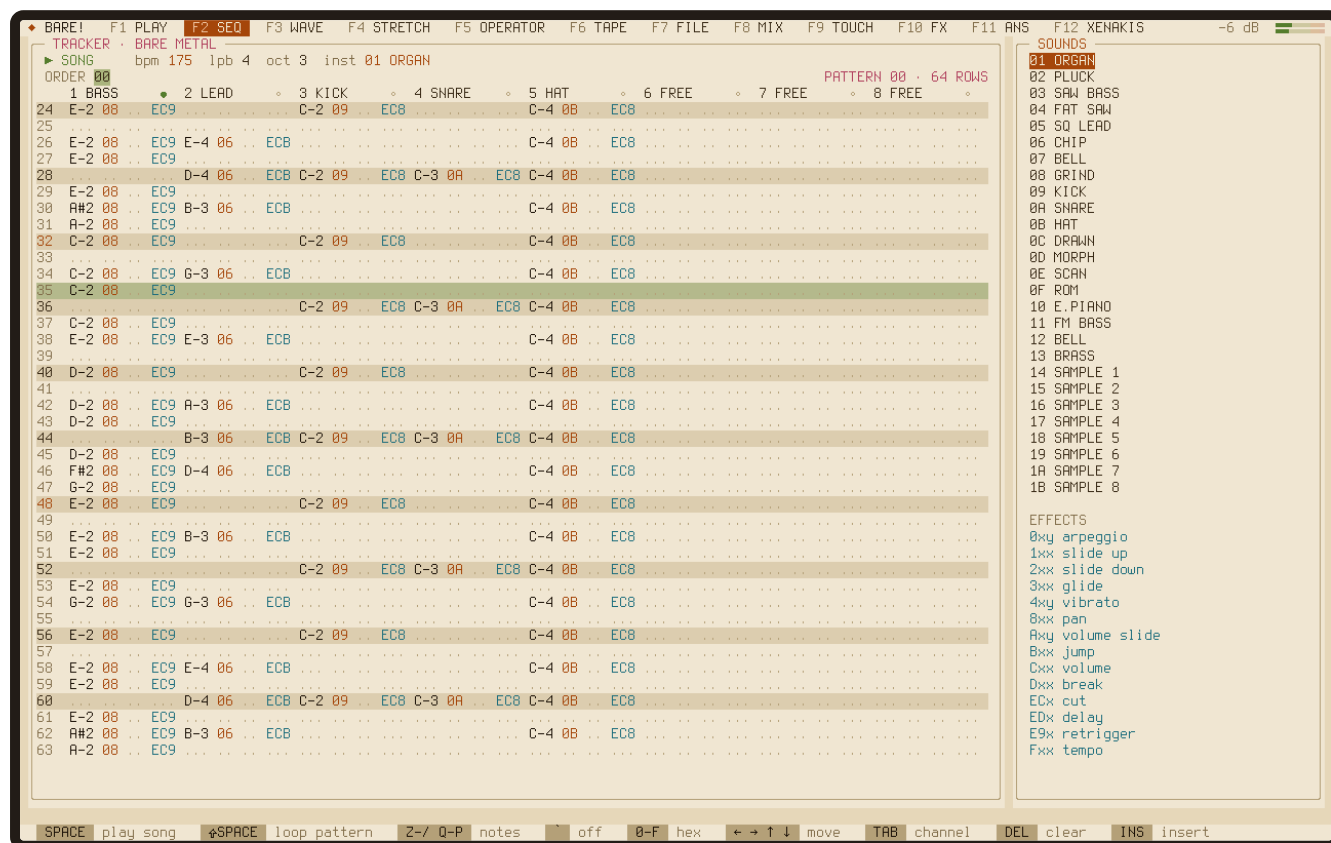
YOUR OWN INSTRUMENTS



Your own instruments

PLAY's **[F1]** again steps from the Omnichord to instruments made from text files, and the tab takes the instrument's name. Five come with it: STYLO (the touchpad as a Stylophone's strip), THEREMIN (pitch without steps), GRIDPADS (a grid of pads with an arpeggio), SIEVHARP (a harp tuned to a sieve) and SHRUTI (a shruti box, the drone box of Indian music: the keys open and close its reeds, tuned in just intonation from Sa, and a finger moving to and fro on the touchpad pumps its bellows; **[Enter]** held pumps too). Anyone can write more: put the file in the stick's **INSTR** folder; **INSTRUMENTS.md** has the format. On an instrument's page, **[↑]** **[↓]** **[←]** **[→]** turn its knobs, **[Space]** holds what is played, the letter rows and MIDI play it; its sound is also one of the sounds everywhere else, by its name.

A tracker. Rows run down the page and each of the 8 channels is a column: note, instrument (any sound, the samples included), volume and effect. A channel plays one note at a time and holds it until its next note or a note-off. A pattern is 16 to 64 rows; the song is the order list at the top. It plays on while you use the other pages, and every row and effect lands on the exact sample.



F2 · SEQ

KEYS

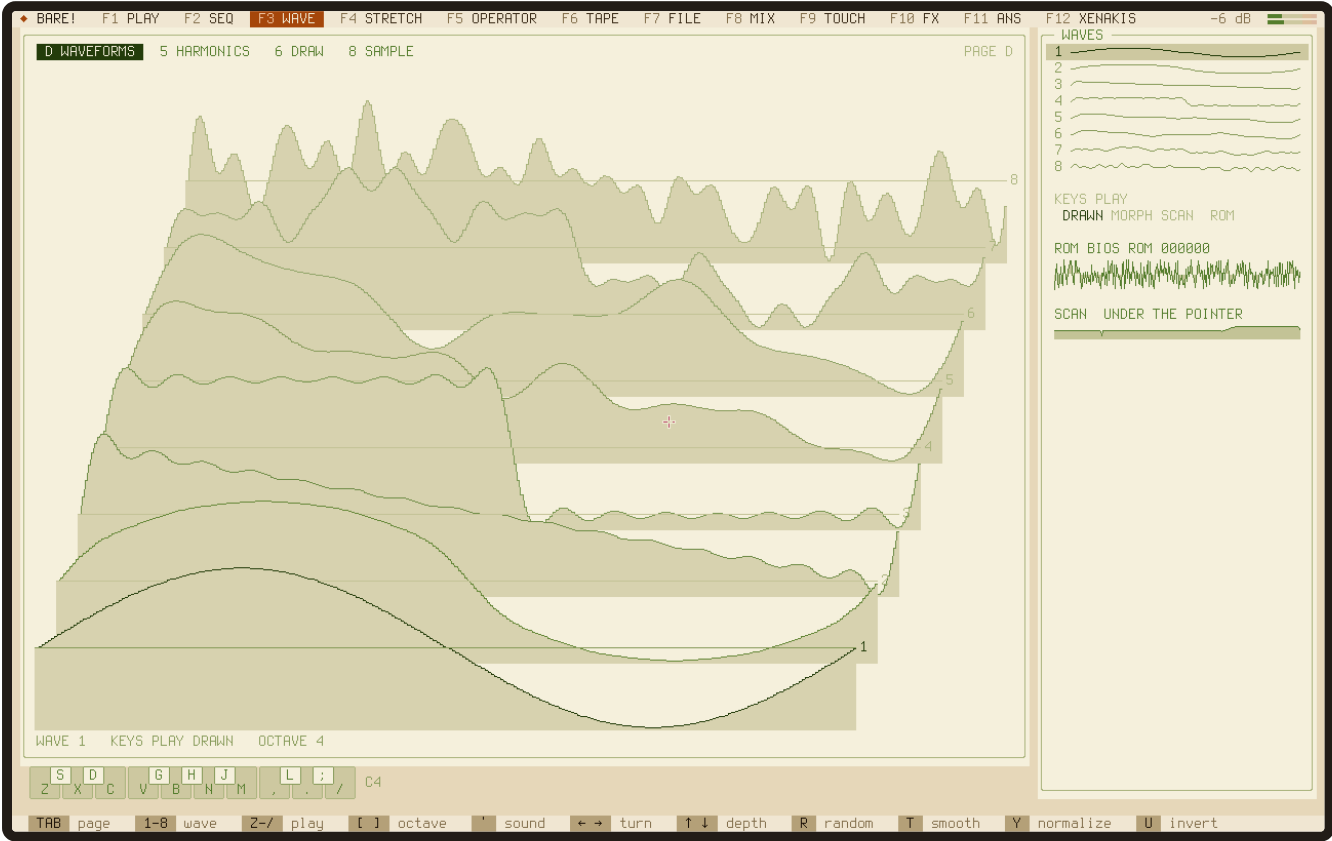
Space / Shift + Space	play the song from the order entry / loop the pattern; stop
Z ... / and Q ... P	notes, two octaves of piano keys (in the note column), with the current sound
`	note-off
0 - 9 A - F	hex, in the instrument, volume and effect columns
arrows, Tab , PgUp , PgDn , Home	move (or click a cell)
End	
Del / Ins	clear the field / push the channel down a row
[] / Shift + []	octave / the sound new notes get
- =	tempo; Shift + P rows a beat
Shift + PgUp PgDn	the pattern on screen; Shift + L its length
\	the order list: ← → entry, ↑ ↓ its pattern, Ins repeat, Del remove, Enter back
Shift + C Shift + V	copy / paste a pattern

KEYS

Enter	mute the channel
Shift + 1 2 3	load demo: BARE METAL · HARBOR · EMPTY (and play it)

Effects, 16 ticks to a row: **0xy** arpeggio (+x, +y semitones), **1xx** / **2xx** slide up/down, **3xx** glide to the note, **4xy** vibrato, **8xx** pan (80 centre), **Axy** volume up x or down y, **Bxx** jump to order entry xx, **Cxx** volume, **Dxx** next pattern from row xx, **ECx** cut after x ticks, **EDx** delay x ticks, **E9x** retrigger every x ticks, **Fxx** tempo (20–FF) or rows a beat (01–1F). **00** repeats an effect's last value. On wide screens the list of sounds and effects sits beside the pattern. Songs from 1.0 load into pattern 0, their ties held and their gates turned into cuts. Both demo songs are original.

A Fairlight-style page, green on black, in four sub-pages: `Tab` / `Shift+Tab` or a click on the tabs. The bottom two letter rows are a piano keyboard for the current sound (`Z` C, `S` C#, `X` D ... `/` E; `[` `]` octave; `Space` plays it at its own pitch). Here the touchpad is a pen: the pad stands for the screen, and a finger draws, drags and picks where it lands. A mouse does the same with the button held.



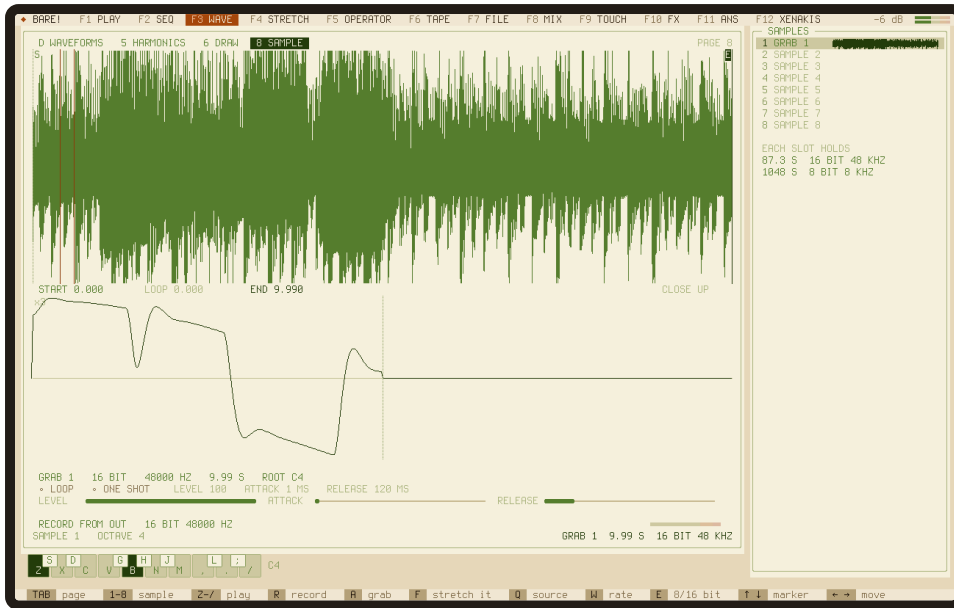
F3 · WAVE – the eight waves in 3D

PAGE

D WAVEFORMS	the eight wave slots stacked in 3D, front to back: the path MORPH plays through. <code>1</code> – <code>8</code> or a click picks a slot; <code>←→</code> turn, <code>↑↓</code> depth
5 HARMONICS	32 harmonic bars build the slot's wave: draw them, or <code>←→</code> pick one and <code>↑↓</code> set it
6 DRAW	draw one cycle of the slot's wave; <code>←→</code> cursor, <code>↑↓</code> nudge 4 samples
8 SAMPLE	the sampler, below

On the wave pages: `R` random, `T` smooth, `Y` normalize, `U` invert, `I` clear, `O`/`P` copy/paste. `'` chooses what the keyboard plays: DRAWN (the slot), MORPH (from the slot to slot 8 over 1.5 s), SCAN (the screen under the pointer is the wavetable) or ROM (this machine's BIOS, flash, kernel or low memory as a wavetable: `\` source, `-` `=` and `PgUp` `PgDn` move through it; every machine sounds different).

SAMPLER



Sampler

Eight sample slots, sized from the machine's memory: 8 s each at full quality on a 32 MB PC, 87 s with 512 MB or more.

- **[R]** arms a recording: it starts when the sound does and stops at **[R]** or when the slot is full. **[Q]** picks what it records: OUT (everything you hear), FREEZE (the stretcher alone) or IN (the line in or microphone; it turns the first input on if none is).
- **[A]** grabs the phrase you just played from the stretcher's last 10 seconds. Nothing needs arming first.
- **[E]** 8 or 16 bit and **[W]** the rate (48, 32, 24, 16, 8 kHz) convert the slot and set the format of the next recording. 8-bit frames play without smoothing, as on the old samplers; that is where their grit comes from.
- **[↑↓]** picks a marker (start, loop, end), **[↔]** and **[PgUp]** **[PgDn]** move it, or drag it. **[K]** loop on/off. Under the sample is a close-up of the frames at the marker, or of the loop's splice.
- **[□]** **[=]** root note, **[']** one-shot, **[T]** trim to the markers, **[Y]** normalize, **[U]** reverse, **[I]** clear, **[O]**/**[P]** copy/paste; level, attack and release are the sliders.
- **[F]** hands the sample to the stretcher, which freezes on it (**[F4]** to shape it).

Elsewhere the slots are the sounds SAMPLE 1–8: chord and strum sounds, tracker instruments. They are saved with the project, frames included.

The last 10 s of everything you played is always being captured. **[Space]** (or **[Shift]+F** from anywhere) freezes it: the stretcher finds the last phrase you played (back to the last 0.4 s of quiet) and paulstretches it, looping through it, in stereo. **[←→]** slower/faster $\times 1 \dots \times 1024$, **[↑↓]** window, **[L]** **[J]** scrub, **[=]** mix, **[Enter]** stay on one spot (infinite). Chords and strums still play on top.



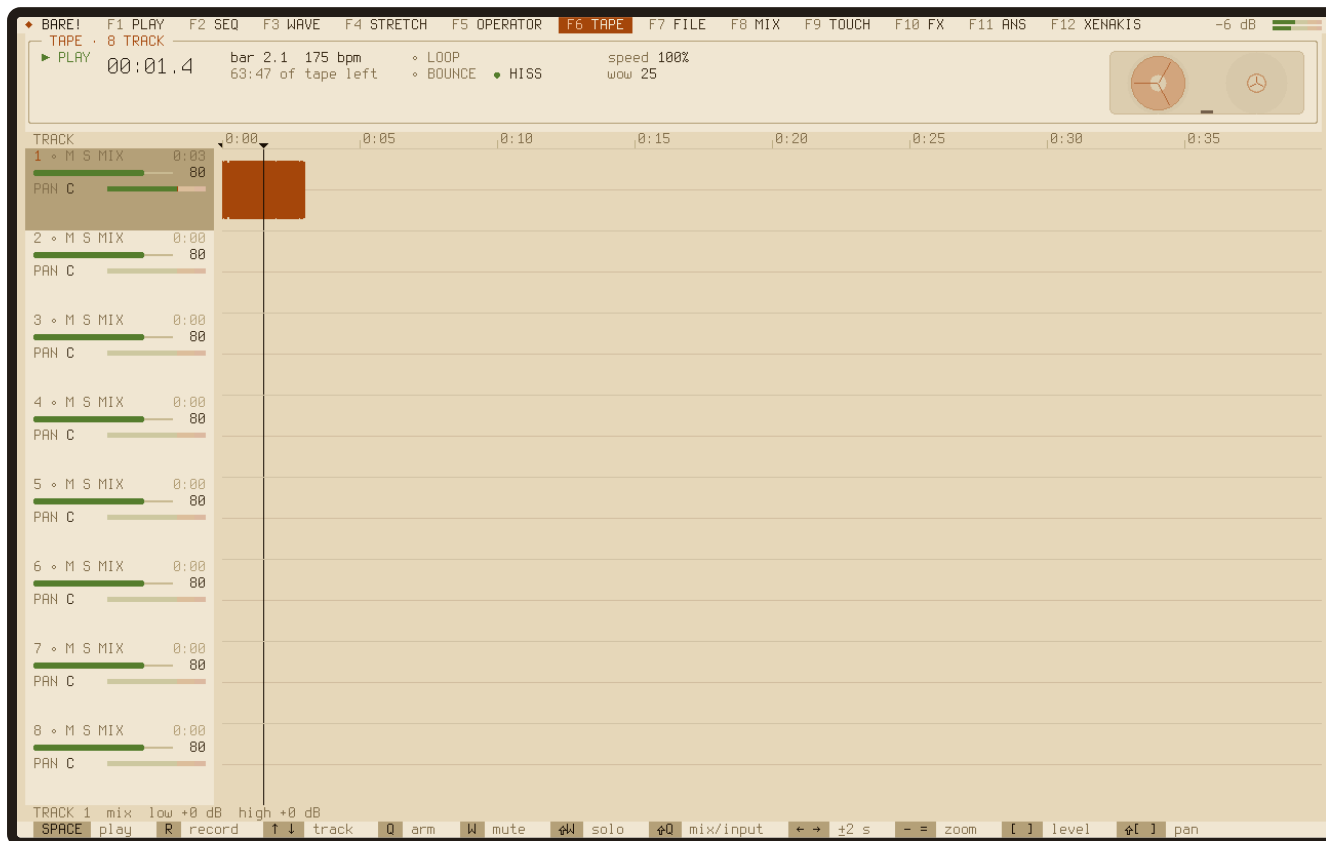
4-operator FM, Ableton-Operator style. 4 patches (**Home**/**End**), 8 algorithms (**PgUp**/**PgDn**): serial, Y, two-pair, fan-in, 2+2, 3+1, 1-to-3, additive—drawn as a diagram), op-4 feedback (**-**/**=**). Per operator: wave (sine/tri/saw/square), ratio 0.5–16 + fine, level, a full ADSR—the carrier envelopes are the amp envelope, so notes release properly—velocity (how much softer a soft note makes it; on a modulator, darker) and key scaling (quieter up the keyboard, down to –12 dB four octaves above C3). Per patch, an LFO: rate 0.1–20 Hz, wave (sine, triangle, saw, square, random), and how far it moves the pitch (up to a semitone), the level (tremolo) and the modulators (the timbre); each note has its own. **Tab** picks the operator, **↑↓** the parameter, **←→** / **[]** adjust; or press on a parameter and drag across its bar. **Enter** renames the patch. Patches save with the project and are the FM 1–4 sounds everywhere (chord/strum sounds on PLAY, track sounds in the sequencer).



F5 • OPERATOR

Strum-plate keys release sustaining sounds on key-up; one-shot sounds (PLUCK, BELL, drums) ring out.

An 8-track recorder. The tracks are lanes of waveform under a ruler, as in a DAW, and it still has a cassette's character: reels, wow and flutter, hiss, varispeed, saturation, bounce.

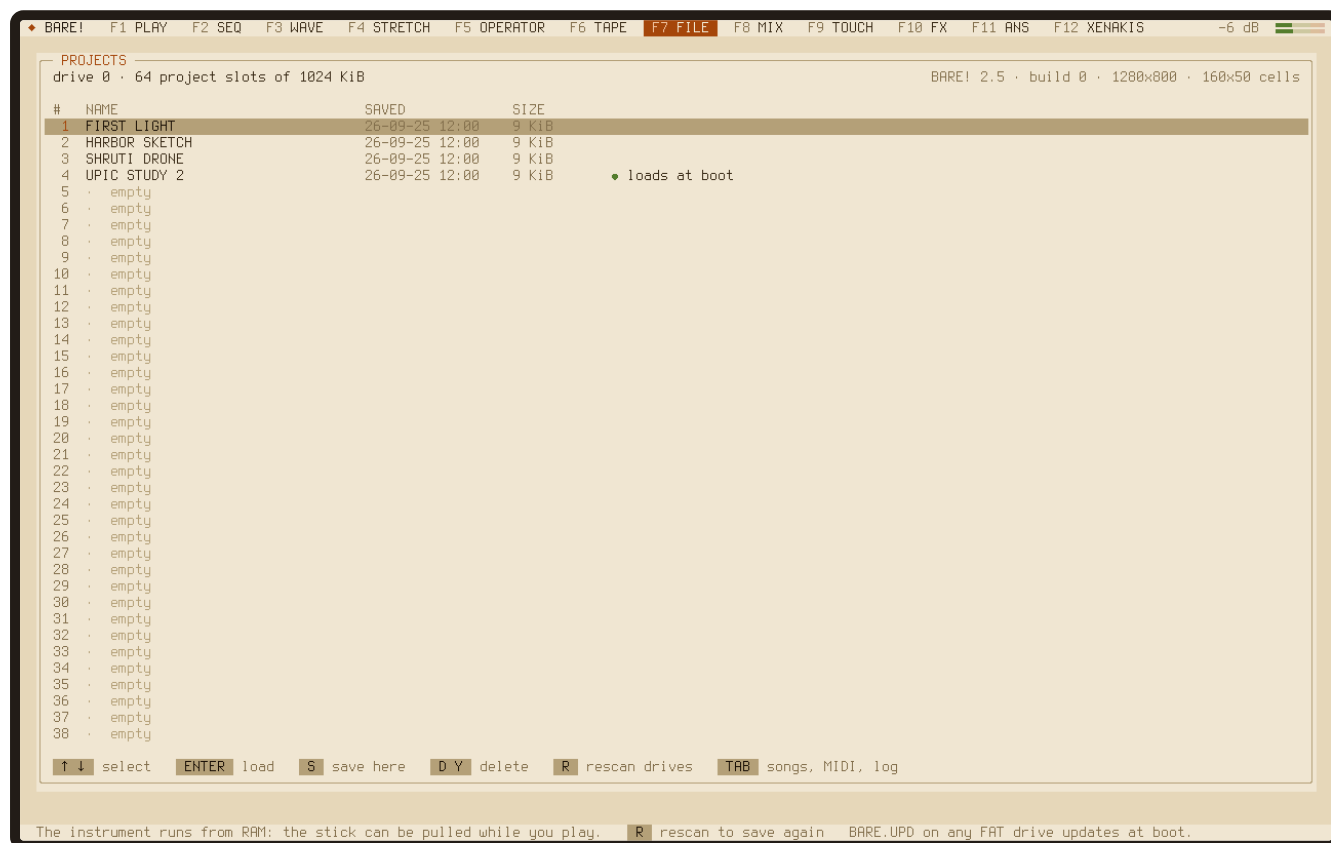


F6 · TAPE

- **[↑][↓]** (or click a lane) picks a track, **[Q]** arms it, **[R]** records on the armed tracks (again to punch out), **[Space]** plays and stops. Recording replaces what was on the track. **[↑][Q]** switches a track between MIX (what you hear: chords, strums, sequencer, stretcher, the input when MON is on) and IN (the line in or microphone alone, heard or not).
- What you play while the tape rolls lands where you heard the tape, not later: the sound card's delay is taken off.
- Per track: **[L][J]** level, **[↑][L][J]** pan, **[↑][,][↑].** low shelf and **[↑];[↑']** high shelf (± 12 dB), **[W]** mute, **[↑][W]** solo, **[E]** twice erases it.
- Moving: **[←][→]** ± 2 s, **[PgUp][PgDn]** ± 15 s, **[Home]/[End]**, a click in a lane or on the overview under the lanes; **[−][=]** zoom (5 s to 20 minutes across).
- A region: drag across the ruler, or **[↑][I]** and **[↑][O]** at the head. **[']** loops it (without a region, the loop is the whole song). **[↑][C]** copies the region of the track, **[↑][V]** pastes it at the head on the selected track, **[↑][D]** erases the region on the track. Record over a loop with BOUNCE on and it becomes a loop pedal with overdubs.
- **[T]** BOUNCE (the recording takes the other tracks' playback too), **[Y]** hiss, **[U]** wow and flutter, **[I]/[O]** varispeed ($\frac{1}{2}\times$ to $2\times$, playback only; **[P]** back to normal).

The sound lives in RAM, in blocks taken as tracks are recorded, so the memory goes to whichever tracks you use: about 3 minutes of one track's sound in all on a 32 MB PC, over an hour with 512 MB. The timeline is 30 minutes long. The TAPE channel on the MIX page is the tape's output. Saving a project saves the tape and the samples with it; if the stick is too full for the tape, the project is saved without it and says so.

Projects on the stick: **[↑↓]** select (or click), **[Enter]** load, **[S]** save (type a name, **[Enter]**), **[D]** then **[Y]** deletes. Saves are dated from the machine's clock. The last saved/loaded project comes back at the next boot. **[Tab]** steps through SONGS, MIDI and the log: the kernel log and the state of the sound device (a photo of it helps with a problem on a machine without a serial port); there **[T]** plays a test tone, **[P]** turns every output on, ignoring the jacks, and **[W]** writes a hardware report to the stick: HB-LOG.TXT (the whole log), HB-PCI.TXT (every PCI device) and HB-ACPI.BIN (the firmware's ACPI tables), for looking at a machine from another computer.



F7 · FILE

SONGS

[E] renders the song into a WAV on the stick's FAT partition, the part any computer reads: the tracker from its first order entry and the tape from its start, 48 kHz 16-bit stereo, as SONG0001.WAV and on, faster than real time (the speakers are quiet meanwhile; **[Esc]** stops). The WAVs in the stick's root are listed: **[S]** brings the selected one into a sample slot (**[I][J]** picks it), **[T]** onto the tape's selected track at the head. Imports take PCM WAVs, 8, 16 or 24 bit, mono or stereo, any rate.

MIDI AND SYNC

A USB MIDI keyboard, controller or interface, a module on a serial port, or an MPU-401 (a sound card's game port). USB MIDI devices can be plugged in any time while our USB stack runs: always on a UEFI boot; on a BIOS boot **[U]** here takes USB over from the BIOS until the next boot (the stick then goes through our driver too). Serial ports run at 38400 baud, the "PC" setting of Roland and Yamaha modules and keyboards. **[←][→]** picks the port, **[↑][↓]** the channel it listens to.

- In: notes play the page's sound with their velocity (on SEQ, stopped, they go in at the cursor), the sustain pedal holds, pitch bend is ± 2 semitones, the mod wheel opens the filter, program change picks the strings' sound. **[C]**: an incoming

clock sets the tempo, and start and stop run the tracker.

- Out: **[N]** the tracker's channels 1–8 as MIDI channels 1–8, **[K]** clock, start and stop at the tempo, **[T]** thru.
- The last messages in are listed, so you can see what a keyboard sends. The settings stay on the stick.
- **[L]** turns on Ableton Link: tempo, beat and start/stop shared with other programs on the same network (Live, most music apps on phones and computers, and another computer running this). The tracker and the rhythm section play in time with them; a start comes in on the next bar. The tempo keys change everyone's tempo.

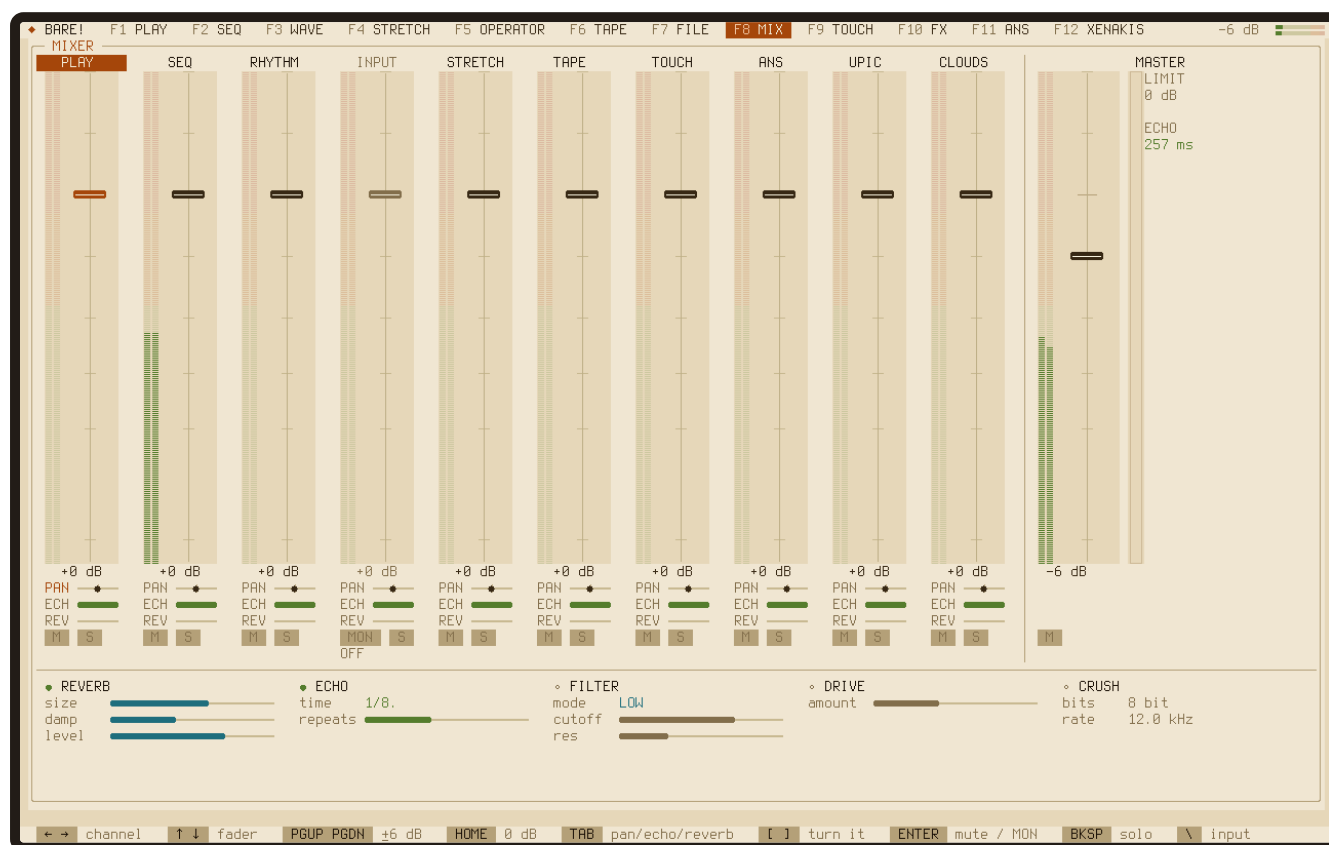
NETWORK

A cable between two computers running this, or into a router, is enough: the address comes from the router (DHCP), or without one each picks its own (169.254.x.y). The line under LINK shows it; the instrument answers ping. It works with the Ethernet ports of Intel's chips (e1000 family, which includes most ThinkPads), and with USB Ethernet adapters that follow the standard (CDC-ECM or CDC-NCM: most USB-C ones made for Macs and iPads). Wi-Fi does not work.

INSTALL

Puts the instrument on the computer's own disk, so it starts without the stick. The list shows the other drives with what is on them now ("GPT, 3 partitions" is usually another system). **[↑↓]** picks one, **[I]** asks, and only after you type **ERASE** and press **[Enter]** is the drive erased and written: the stick's boot partition, then its projects. Then take the stick out and restart; if the computer still starts its old system, choose the disk in its boot menu. Updates (BARE.UPD) and saving work as on the stick, and with the stick plugged in too, the disk the computer started from keeps the projects. On a BIOS boot any disk the BIOS sees can take it; on a UEFI boot, SATA and NVMe disks (also behind Intel VMD).

A strip per source: PLAY (chords, strums, the WAVE keyboard, the instruments), SEQ, RHYTHM, INPUT, STRETCH, TAPE, TOUCH, ANS, UPIC and CLOUDS (and DOOM, once Doom has run), each with a meter, fader (−60 to +12 dB), pan, echo and reverb sends, mute and solo; then the master with the volume and how much the limiter is taking off; and the effects along the bottom.



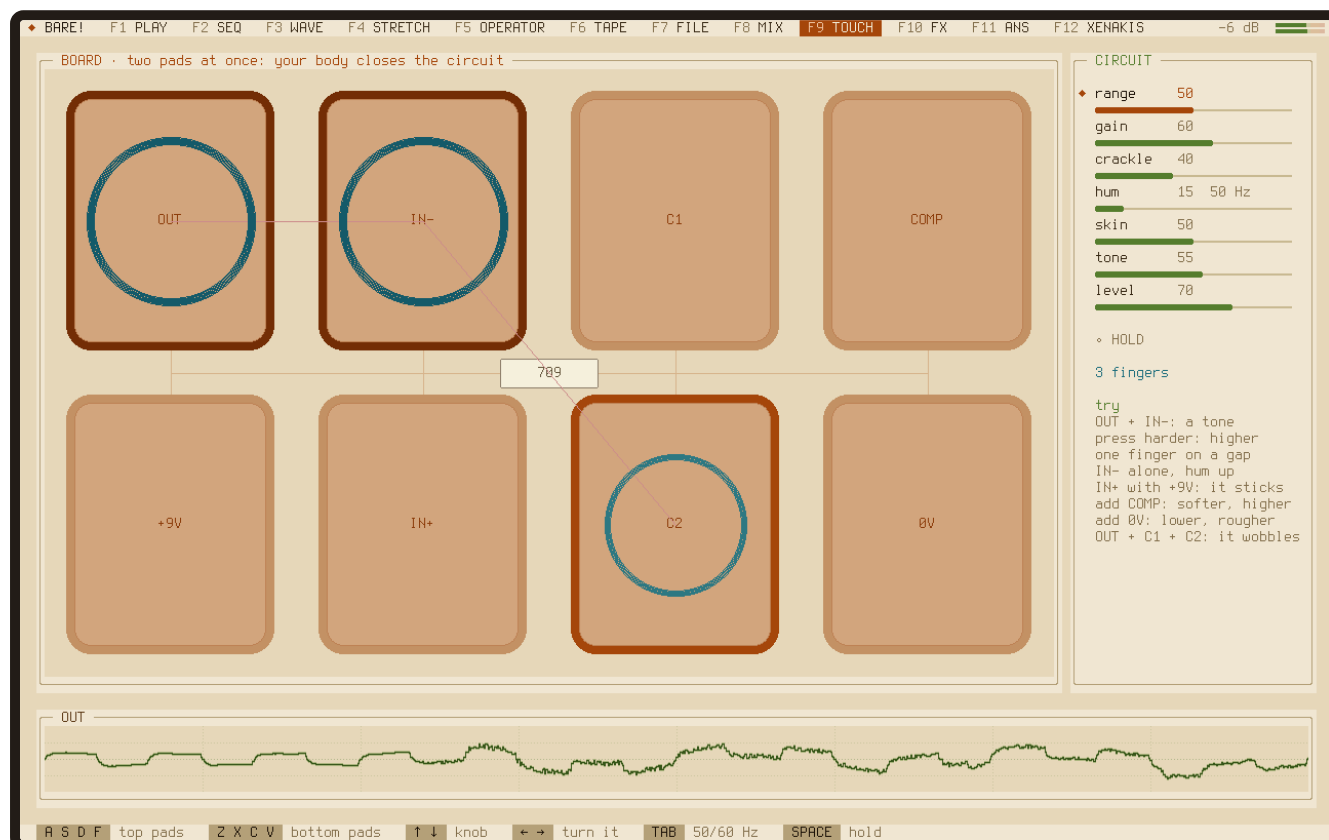
F8 · MIX

- $\leftarrow$ $\rightarrow$ channel, $\uparrow$ $\downarrow$ fader (PgUp PgDn 6 dB, Home 0 dB, End off), Tab chooses pan, echo send or reverb send and [] turns it, Enter mute, Backspace solo; or drag the faders, pans and sends and click M and S.
- Past the master, $\leftarrow$ $\rightarrow$ reaches the effects: REVERB (size, damping, level), ECHO (time as a note length at the tempo, repeats), and on the master FILTER (low, band or high pass, cutoff, resonance), DRIVE and CRUSH (bits, and the rate the held frames leave). $\uparrow$ $\downarrow$ picks a setting, [] changes it, Enter turns the unit on or off.
- The chord buttons and strum keys still play here. The mixer and the effects are saved with the project.

INPUT

The line in or a microphone, on the MIX page's INPUT strip: [] steps through what the sound chip has (LINE IN, MIC, BUILT-IN MIC ... then off), [] sums it to mono. It is heard only with MON on (Enter on the strip), since a laptop's own microphone next to its speakers howls; use headphones to listen while you play. What is heard goes where everything else does (the stretcher, the tape's MIX tracks). Unheard, it can still be recorded: by the sampler (source IN) and by a tape track set to IN. A microphone plugged into a jack gets its bias voltage and 20 dB of boost where the chip has them.

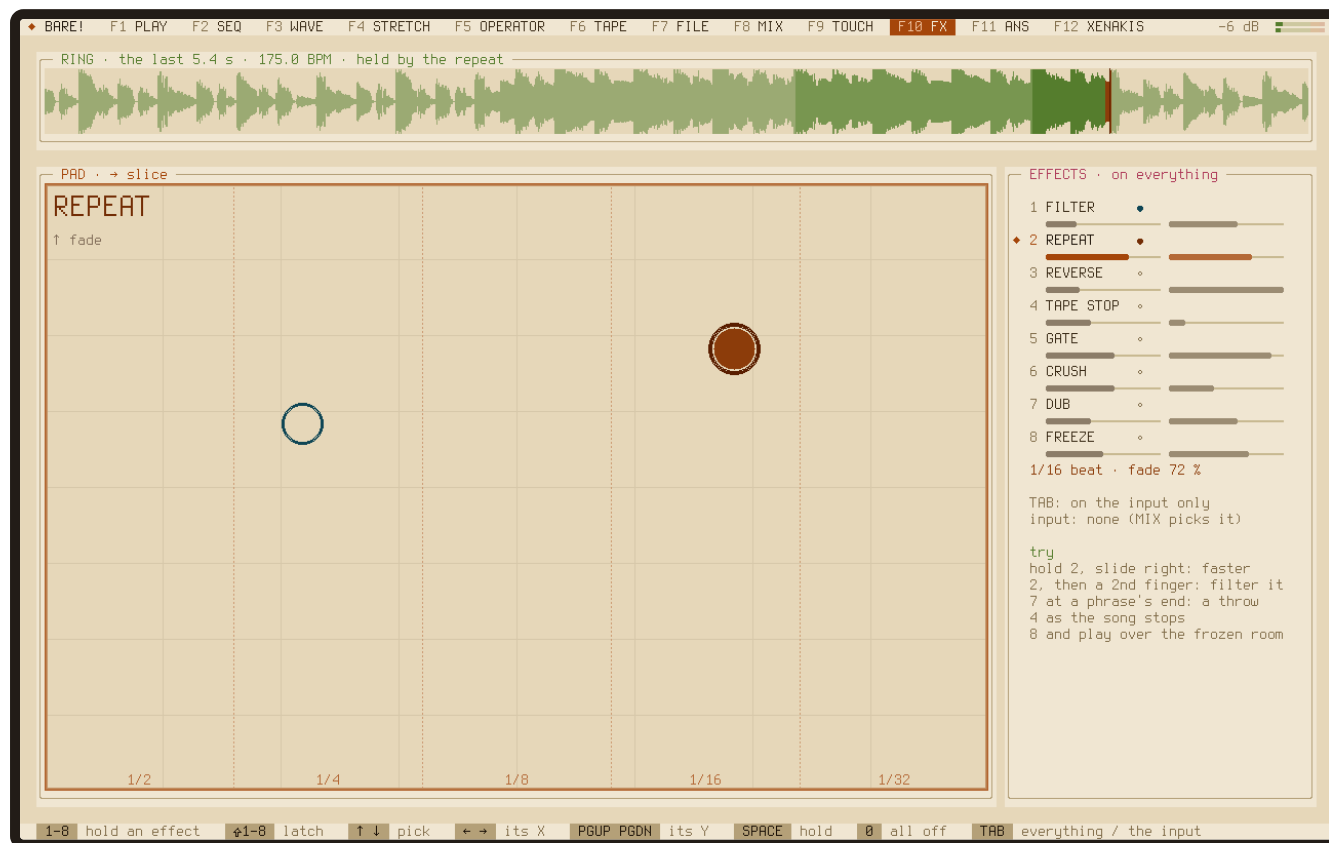
A small circuit you play with your fingers, like a crackle box. Eight bare pads lead into it: an op-amp's output (OUT) and inputs (IN–, IN+), a capacitor behind each input (C1, C2), its compensation point (COMP), the supply (+9V) and ground (0V). A finger on a pad joins it to your body; two pads touched at once are joined through your body, and the circuit starts to sound. Pressing harder lowers the skin's resistance and raises the pitch; a light touch crackles; one finger on the gap between two pads joins them directly; a finger alone picks up the mains hum (turn HUM up).



F9 • TOUCH

- Touchpad: every finger is a finger on the board (the pad stands for the board), with its pressure where the pad reports one. Mouse: click and hold on the board (it presses harder the longer you hold; the right button presses hard at once).
- **A S D F** / **Z X C V**: a finger on each pad of the top and bottom rows; a key held longer presses harder, **Shift** presses hard at once. **Space** holds the key fingers down after the keys go up (press a key again to lift it).
- MIDI notes put a finger on pad (note mod 8), as firm as the velocity.
- **↑ ↓** picks a knob, **← →** turns it (or drag its bar): RANGE (the capacitors: the pitch range), GAIN (the op-amp's: low is soft and round, high is square), CRACKLE (how often a light contact breaks), HUM, SKIN (dry to moist), TONE and LEVEL. **Tab** switches the hum between 50 and 60 Hz. The knobs are saved with the project; the sound has its own channel on the MIX page.

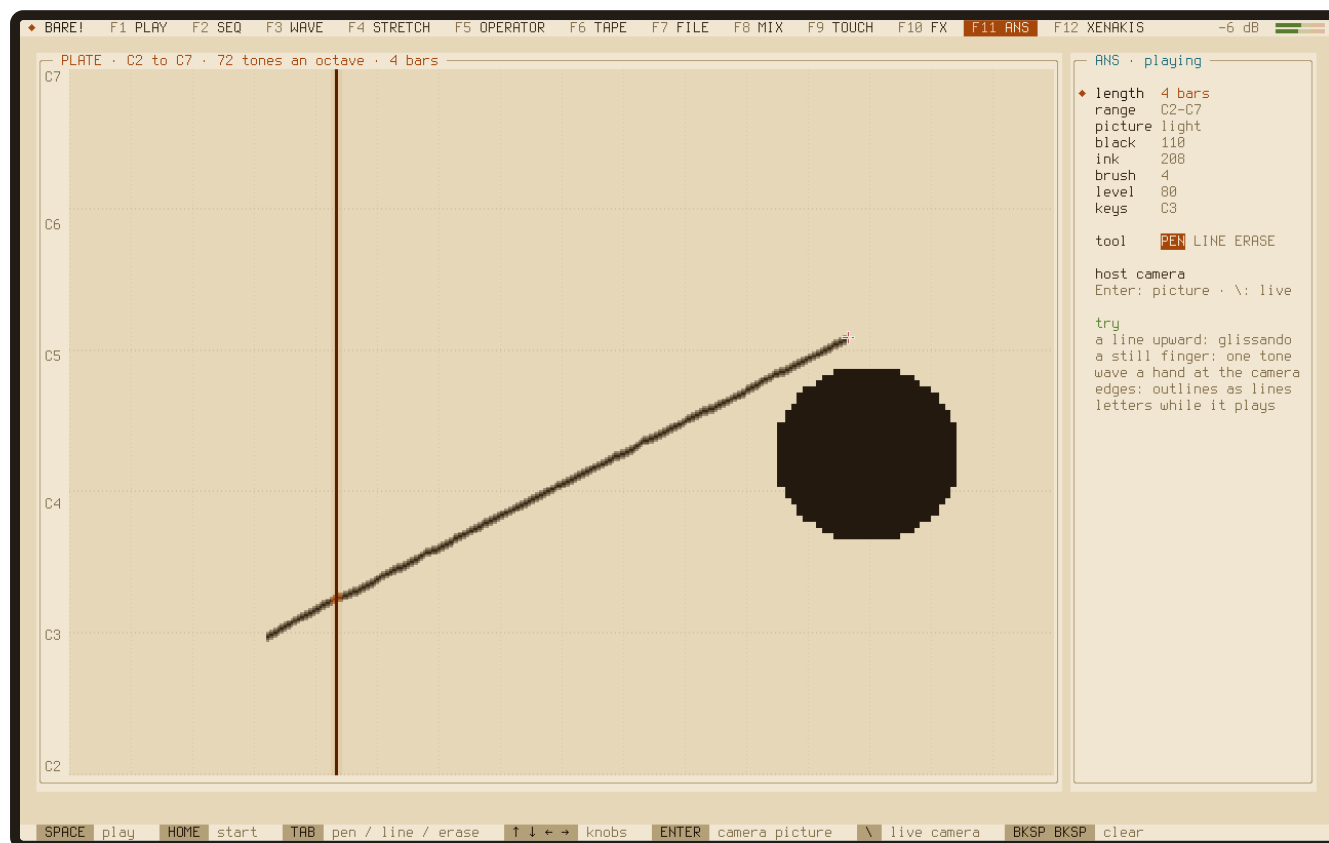
Effects to play live, on everything you hear or only on the input (a guitar or a microphone through the line in: pick it on the MIX page, then **[Tab]** here). Start a song or the rhythm, then:



F10 · FX

- **[1]–[8]** hold an effect while the key is down; **[Shift]+[1]–[8]** latch one on or off; **[Space]** latches whatever is sounding (again: lets all latches go); **[0]** turns everything off.
- The pad is the effect you picked last (**[↑]** **[↓]**, or click it in the list): the touchpad or the mouse on the pad plays it and sets its two settings, left-right and up-down (**[←]** **[→]** and **[PgUp]** **[PgDn]** do the same from the keys). A second finger plays the filter at the same time.
- **FILTER**: left of the middle a low-pass closing down, right of it a high-pass opening up, nothing in the middle; up is resonance. **REPEAT**: loops the last slice (1/2 to 1/32 of a beat, left to right: slide right while holding it and the stutter speeds up); up fades it. **REVERSE**: plays backwards from the moment you press (1/4 to 2 beats); up is how much. **TAPE STOP**: slows down to a stop (in 0.1 to 2 s); up adds wobble. **GATE**: chops in time (1/4 to 1/32 of a beat); up is how deep. **CRUSH**: fewer bits to the right, a lower rate upward. **DUB**: throws everything into the echo while held, the echoes ring on after (left-right the echo's time, up the repeats). **FREEZE**: the same into a reverb as long as a room can be (dark to bright, up is its level).
- Lengths follow the tempo (the song's, or the Link session's). The ring along the top holds the last seconds the effects read from: the write head moves along it and stops while a repeat holds it.
- MIDI: notes 36–43 (the first row of a drum pad) hold effects 1–8; CC 16 and 17 are the pad's two settings; CC 102–109 hold effects 1–8 while at 64 or more.

After Evgeny Murzin's ANS synthesizer (1957), which played light through scratches in a glass plate. The plate here is 360 pure tones, 72 to the octave (a sixth of a semitone apart), over five octaves up the page, and time across it; a slit crosses it in a number of bars of the tempo (or of seconds) and plays every tone it lights, as loud as the plate is bright there.

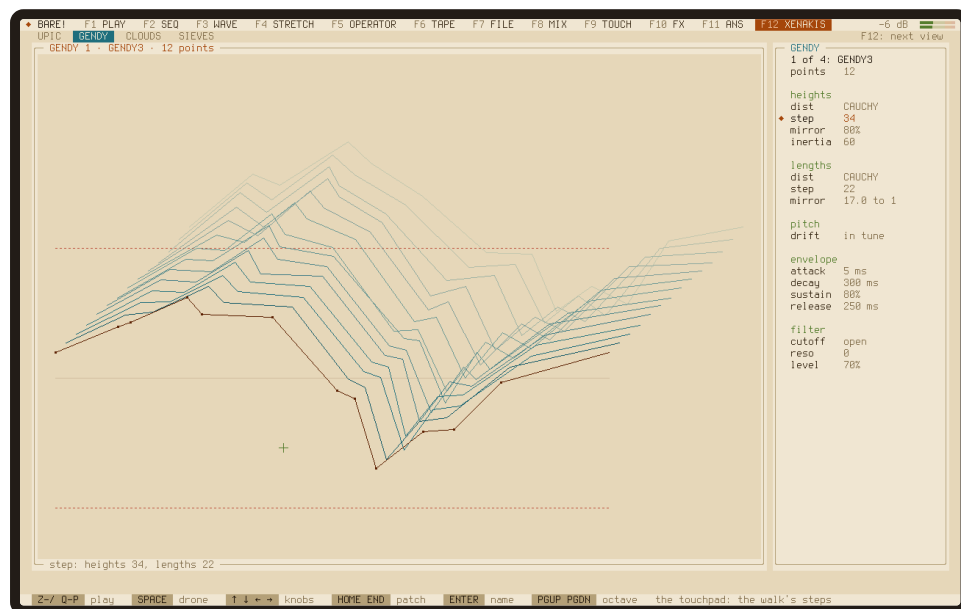


F11 · ANS

- Draw on it: every finger on the touchpad scratches (harder is brighter where the pad can tell), the mouse draws, its right button erases. **[Tab]**: pen, line (press, drag, let go: a straight glissando), eraser.
- **[Space]** plays, **[Home]** goes back to the start. **[Backspace]** twice clears the plate (**[Ctrl]+[Z]** brings it back).
- The camera (a USB webcam, or the laptop's own): **[Enter]** lays its next picture over the plate—the picture's light above the BLACK level, or its edges (PICTURE: light / edges)—and switches the camera off again; **[V]** lets the live camera be the plate while it plays: whatever the camera sees, you hear. On a BIOS (legacy) boot the BIOS keeps USB: pressing either key twice hands USB to BARE! until the next boot (as **[U]** in the MIDI view does), then the camera is found.
- The letter rows are a keyboard of its tones (**[Z]-[I]**, **[Q]-[P]**, two octaves from KEYS; **[PgUp]** **[PgDn]** move it), and so are MIDI notes: held while the plate plays, they write their row under the slit.
- **[↑]** **[↓]** picks a knob, **[←]** **[→]** turns it: LENGTH (1–32 bars, or 2–60 s), RANGE (where the five octaves start: C1 to C4), PICTURE, BLACK, INK (how bright the pen draws), BRUSH, LEVEL, KEYS. The plate is saved with the project; the sound has its own channel on the MIX page.

Four of Iannis Xenakis's ways of making music, a view each: UPIC, GENDY, CLOUDS, SIEVES. **F12** again goes to the next view; a click on the bar under the title picks one. Clouds and UPIC have their own channels on the MIX page, and everything here is saved with the project.

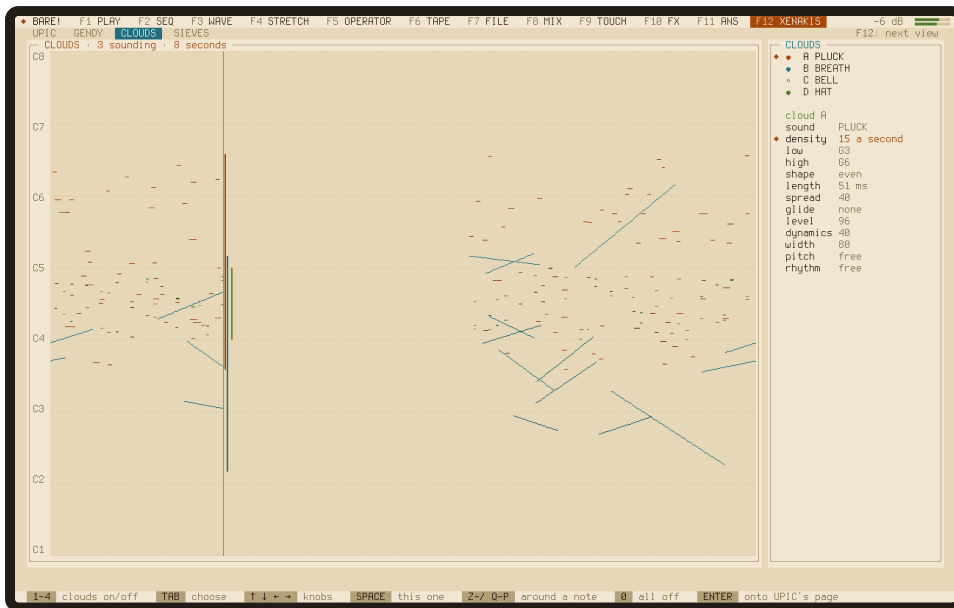
GENDY



GENDY

His dynamic stochastic synthesis (GENDY3, 1991): a wave whose corners take random steps every period and bounce off mirrors. Four patches, played anywhere as the sounds GENDY 1–4 (strum plate, tracker, MIDI). The picture is the wave walking, the last periods behind it. The letter rows play it (**PgUp** **PgDn** octave), **Space** holds a drone, the touchpad (or the mouse on the picture) sets the step sizes: across for the heights, up for the lengths. **↑** **↓** **←** **→**: POINTS; the heights' DIST (LINEAR, CAUCHY, LOGIST, HYPCOS, ARCSIN, EXPON), STEP, MIRROR and INERTIA (the steps walk too: drift instead of jitter); the lengths' DIST, STEP and MIRROR; DRIFT (the pitch walks as well); envelope, filter, level. **Home** **End** patch, **Enter** renames it.

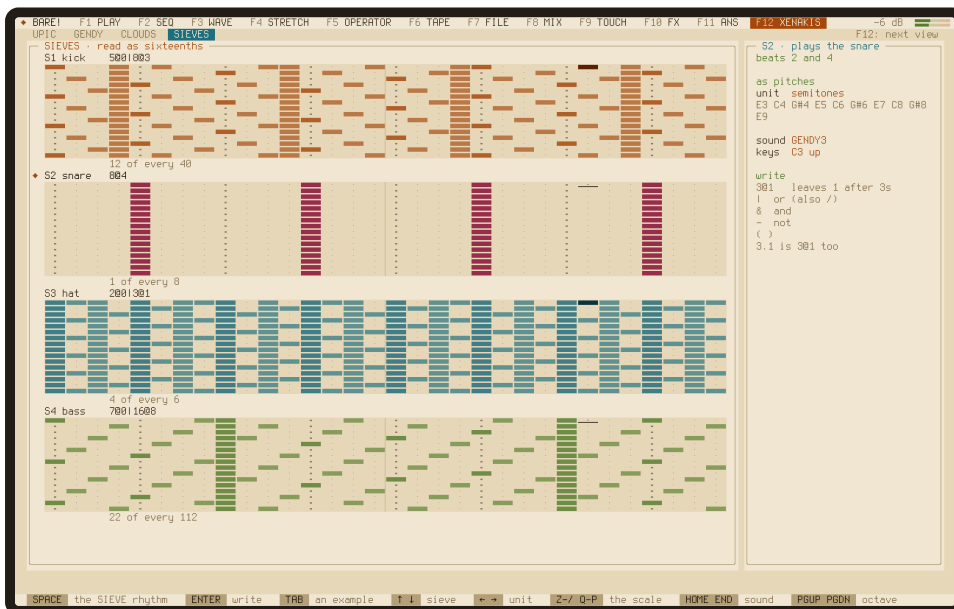
CLOUDS



CLOUDS

Masses of notes set by probabilities, after Pithoprakta and Achorriopsis: so many a second at random moments, spread over a band, each note gliding at its own speed. Four clouds, each with its own sound. **[1]–[4]** switch them, **[0]** stops all, **[Tab]** picks one for the knobs (sound, density, band, shape, length and its spread, glide, level, dynamics, width, a pitch sieve, a rhythm sieve). Each finger on the touchpad plays a cloud while it stays down (up and down moves the band, across the density); the mouse does it for the chosen one, and the letter rows and MIDI play it around the note held. **[Enter]** writes the chosen cloud onto UPIC's page. The picture is the score as it is written.

SIEVES

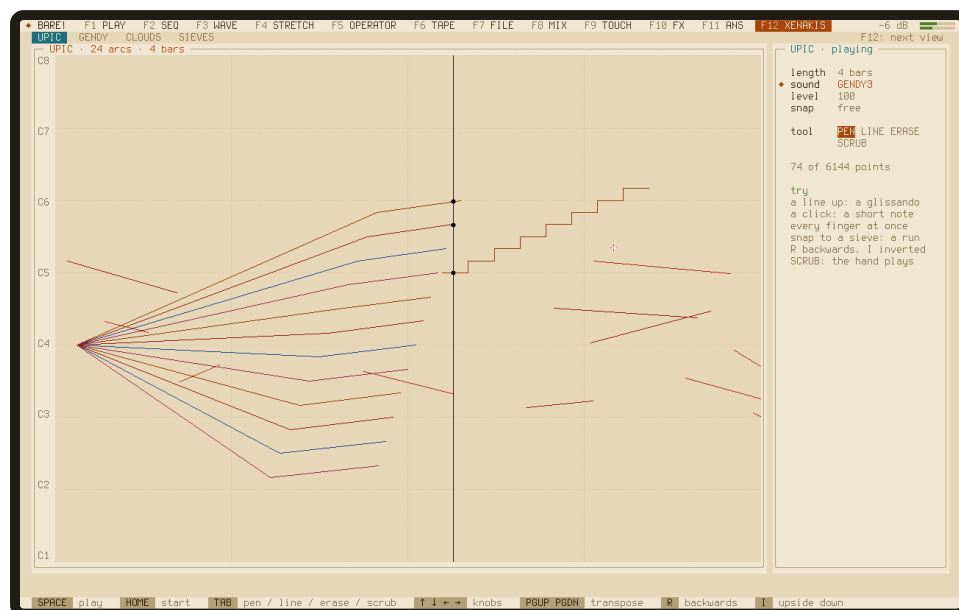


SIEVES

Scales and rhythms from residue classes: **301** is every number that leaves 1 when divided by 3 (1, 4, 7 ...); **[1]** or **[&]** and, **[]** not, and parentheses. Four sieves, drawn as grids of sixteenths, a row a bar. The rhythm section's SIEVE pattern (on PLAY, or **[Space]** here) plays S1 on the kick, S2 the snare, S3 the hat and S4 the bass, counted from the start, so a sieve longer than a bar keeps turning across bars. **[Enter]** writes a formula (**[Shift]+[2]** @, **[Shift]+[V]** |, **[Shift]+[7]** &; **[.]** and **[/]**

work too), **Tab** puts in an example, **← →** sets the step read as pitch (a semitone or smaller), and the letter rows and MIDI play the chosen sieve as a scale.

UPIC



UPIC

After the drawing machine Xenakis built in Paris (1977; Mycenae-Alpha): draw arcs of pitch over time, and a cursor plays them, each with the sound chosen when it was drawn. Every finger on the touchpad draws an arc of its own, the mouse draws too, a click is a short note. **Tab**: PEN, LINE, ERASE (or the right button), SCRUB (your hand holds the cursor). **Space** plays, **Home** goes to the start. The knobs: the page's LENGTH (bars or seconds), and the SOUND, LEVEL and SNAP (free, semitones, a sieve) of what you draw next. **PgUp** **PgDn** transpose the page, **R** turns it backwards, **I** upside down, **Backspace** twice clears it.

DOOM

IDDQD

Type `iddqd` on any page and BARE! runs Doom (1993) full screen, with the WAD it finds on the stick. Put one in a `DOOM` folder on the stick's first partition, or in its root: your own `DOOM.WAD` or `DOOM2.WAD`, the shareware `DOOM1.WAD`, or Freedoom's `freedom1.wad` or `freedom2.wad` (free, from freedom.github.io). That partition has room for one WAD of up to about 40 MB. The keys are the 1993 ones: arrows, `[Ctrl]` fires, `[Space]` opens, `[Alt]` strafes, `[Shift]` runs, `[Esc]` is its menu. The F keys go back to BARE! and Doom waits; `iddqd` again returns to it, paused at its menu (typed inside Doom, it is still the cheat). Its sounds and its music are on the mixer's DOOM channel; the music is played by BARE!'s voices (GRIND on the guitars). Saved games go in the same folder. It needs memory to spare: about the WAD's size and 10 MB more, which a 256 MB machine has.



Freedoom, from the stick



E1M1, in BARE!

INSTRUMENTS

WRITE YOUR OWN

An instrument is a small text file. Put it in the `INSTR` folder on the stick (it's on the stick's first partition, the one any computer opens), start BARE!, and it is there: on the PLAY page, press `[F1]` again to step from the Omnichord to each instrument in turn. The file says what it sounds like and how it's played. Nothing in it can crash BARE!: a mistake is shown on the instrument's page and in the log (F7, Tab to LOG), and the rest of the file still counts.

Its sound also becomes a sound like any other: the strum plate, the tracker, clouds and UPIC can all use it by its name.

AN EXAMPLE

```
# STYLO: a pocket organ played with a stylus on a metal strip.

name: STYLO
by: your name
about: a stylus on a metal strip: the touchpad is the strip

[sound]
wave: pulse
width: 30
attack: 1
release: 15
level: 65

[play]
strip: A2 E4
steps: semitones
keys: chromatic A2
mono: yes

[knobs]
knobs: width cutoff vibrato release level
```

`#` starts a comment. Settings are `name: value` (or `name = value`), one a line, in any order within their section. Anything you leave out has a sensible default.

BEFORE THE SECTIONS

SETTING	WHAT IT IS
<code>name</code>	up to 8 letters: the page's tab and the sound's name. Required.
<code>by</code>	who made it
<code>about</code>	one line saying what it is

A file on the stick with the same name as a built-in instrument replaces it.

[SOUND]

SETTING	VALUES	WHAT IT DOES
<code>wave</code>	<code>pulse</code> <code>square</code> <code>saw</code> <code>tri</code> <code>sine</code> <code>noise</code>	the basic waves

SETTING	VALUES	WHAT IT DOES
	<code>drawn</code> <code>morph</code> <code>scan</code>	the WAVE page's drawn waves
	<code>gendy1</code> ... <code>gendy4</code>	GENDY's patches (XENAKIS page)
	<code>fm1</code> ... <code>fm4</code>	the OPERATOR page's FM patches
	<code>sample1</code> ... <code>sample8</code>	the sampler's slots
<code>width</code>	5 – 95	pulse width, % (for <code>pulse</code>)
<code>detune</code>	0 – 100	a second oscillator, a little off: thicker
<code>attack</code> <code>decay</code> <code>release</code>	1 – 8000	milliseconds
<code>sustain</code>	0 – 100	%
<code>cutoff</code>	0 – 127	the filter; 127 is open
<code>resonance</code>	0 – 100	
<code>envelope</code>	0 – 100	how far the envelope opens the filter
<code>vibrato</code>	0 – 100	
<code>level</code>	0 – 100	

[PLAY]

SETTING	VALUES	WHAT IT DOES
<code>keys</code>	<code>chromatic</code> <code>NOTE</code> , <code>scale</code> <code>NOTE</code> , <code>off</code>	the letter rows (Z to / and Q to P) as a keyboard from NOTE: every semitone (the upper row an octave up), or up the scale, key by key (the upper row going on from the lower)
<code>strip</code>	<code>LOW</code> <code>HIGH</code> (two notes)	the touchpad as a strip: across it is the pitch, pressing harder is louder
<code>steps</code>	<code>semitones</code> , <code>scale</code> , <code>free</code>	on the strip: a note a key, the scale's notes, or no steps at all
<code>pads</code>	<code>COLUMNSxROWS</code> <code>NOTE</code> (up to 8x8)	a grid of pads, up the scale from NOTE; the touchpad stands for the grid
<code>scale</code>	<code>chromatic</code> <code>major</code> <code>minor</code> <code>dorian</code> <code>phrygian</code> <code>lydian</code> <code>mixolydian</code> <code>pentatonic</code> <code>minorpenta</code> <code>blues</code> <code>wholetone</code> <code>harmonic</code> , or <code>S1</code> ... <code>S4</code>	the scale; S1–S4 are the sieves on the XENAKIS page
<code>mono</code>	<code>yes</code> / <code>no</code>	one note at a time (the newest)
<code>glide</code>	0 – 2000	milliseconds to slide from note to note
<code>arp</code>	<code>off</code> <code>up</code> <code>down</code> <code>updown</code> <code>random</code>	an arpeggio over the notes held, on the tempo
<code>rate</code>	1 – 8	arpeggio notes a beat
<code>octaves</code>	1 – 4	how many octaves the arpeggio spans
<code>hold</code>	<code>yes</code> / <code>no</code> / <code>toggle</code>	notes stay when let go, until new ones are played (Space turns it on and off); <code>toggle</code> : a key opens its note and the next press closes it, one by one (Space closes them all)

SETTING	VALUES	WHAT IT DOES
tuning	equal, just, just NOTE	equal temperament, or just intonation from a tonic (the keys' first note, or NOTE): each note is the 5-limit ratio of its interval (the fifth 3:2, the fourth 4:3, the major third 5:4 ...)
bellows	yes / no	the notes sound only with air in the bellows: a finger moving to and fro on the touchpad (where it is not a strip or pads), the mouse moving, or Enter held pumps it, and it leaks out over a few seconds. Louder with more air, a few cents flat when it runs low

Notes are written C3, F#2, Bb4 (middle C is C4), or as MIDI numbers.

[KNOBS]

knobs: lists up to 8 settings to turn on the page (↑ ↓ to choose, ← → to turn): any of wave width detune attack decay sustain release cutoff resonance envelope vibrato level glide rate octaves. What you turn is saved with the project.

SHARING

An instrument is one text file, so it can go anywhere text goes: a message, a forum post, a repository. The examples that come with BARE! are in `instruments/` in the source; they're a good place to start copying from.

MACHINES

WHICH COMPUTERS, AND THE STICK

Sound is a fixed-point engine (PolyBLEP pulse and saw, drawn waves, 4-operator FM, noise, samples) with envelopes and per-voice filtering, in stereo; the output is Intel HDA (any PC or Intel Mac since 2005), AC'97 (the PCs before that) or a Sound Blaster 16, and `[Shift]+[B]` switches to 1-bit: the PC speaker's square wave, one note at a time, played through the sound chip (or the real PC speaker on a machine without one).

The instrument runs entirely from RAM: once booted, the stick can be pulled and you keep playing. Saving needs it back in (`[R]` rescans on the FILE page; some BIOSes only re-detect USB after a reboot).

MACHINE	BOOT	SOUND	KEYBOARD	SAVE TO STICK
any PC 1997–2012, BIOS	yes	HDA (2005+), AC97 (1999–2006), Sound Blaster 16	yes	yes
2012–2020 laptop with CSM/legacy boot	yes	yes	yes	yes
UEFI-only (no CSM), e.g. a recent VivoBook	yes (64-bit build)	HDA yes	yes, USB too	yes, through our USB driver
Secure Boot enabled	no—disable it			

Generic setup: **Secure Boot off**; legacy/CSM boot where there is a choice (the BIOS's USB drivers reach more controllers than ours). Machines whose firmware switches off the old PC timer (common on UEFI-only laptops) get their 1 kHz tick from the HPET, or from the CPU's cycle counter with the main loop running the audio. Until the instrument starts, the boot log is drawn on the screen, so a machine that stops during boot shows where.

Touchpads. Two kinds strum the strings and draw with the pen: Synaptics pads on PS/2 (ThinkPads, and most laptops until about 2015) and precision touchpads on I2C (most laptops since). An I2C touchpad is found through the firmware's ACPI tables, switched from its mouse mode to reporting fingers, and asked for a report every 8 ms. The I2C part has been tried on one laptop so far (an ASUS VivoBook, Intel 12th gen): if the pad does nothing on another, the log (`[F7]`, `[Tab]` three times) says how far it got, and `[W]` there writes it to the stick.

USB. On a UEFI boot the instrument drives the USB 3 (xHCI) controllers itself—every PC since about 2012 has one; on Intel 7 to 9 series chipsets it also moves the USB 2 ports over from the older EHCI controller. Sticks, keyboards, mice, tablets and touchscreens (as a pointer), hubs and USB MIDI work, and can be plugged in and out while it plays (`[R]` on the FILE page finds a stick plugged in later). On a BIOS boot the BIOS keeps USB, and its keyboard emulation and disk services go on working; `[U]` in the MIDI view hands USB to our driver until the next boot. There is no driver for the older controllers (UHCI, OHCI, EHCI): on a UEFI-only machine whose stick sits on one of those, the instrument runs without storage.

THE STICK

`bare.img` = MBR + a 64 MiB FAT32 boot partition (Limine, `B00T/HB_A.ELF` and `HB_B.ELF` as two 4 MiB kernel slots, `B00T/H64_A.ELF` and `H64_B.ELF` for the 64-bit kernel UEFI boots, `EFI/B00T/*.EFI`, the instruments in `INSTR`, `LICENSE.TXT` and `NOTICE.TXT`, and the songs you export) + a 128 MiB raw "tape" partition (type 0x7F, magic `HBTAP01`) holding 64 project slots of 1 MiB. On a bigger stick the tape partition is grown to the end of the stick at the first boot (when it is the last partition); a project whose tape and samples don't fit its slot keeps them in a region of that space.

Updating: put `BARE.UPD` (from `make upd`; it carries both kernels) in the root of *any* FAT drive the instrument can see—the stick itself, another stick—and boot. If its build number is newer, each kernel is written into its inactive slot, the CRCs are verified, `limine.conf` is switched to those slots in place, and the machine reboots. A power cut mid-update leaves the old slots bootable. A 1.0 stick, which has one 64-bit kernel, gets its second slot on its first update. Sticks from before the name changed (1.0 to 2.2) look for `HOME BREW.UPD`: the release has the same file under both names. `make usb` re-flashes the boot partition but keeps the tape partition, so projects survive.

Next:

- Hardware controllers made for BARE!: boxes of keys, pads, knobs and encoders laid out for its pages—an omnichord's chord buttons and strum plate, a tracker's step keys, the Etch A Sketch's two knobs—built on RP2040 or ESP32-S3 boards. They talk USB MIDI, which BARE! already understands.
- Musical Etch A Sketch: a page that works like the toy. Two knobs draw one unbroken line, one across and one up and down, and then the drawing plays. It needs two encoders: one of the controllers above, or any MIDI controller with two endless knobs (the arrow keys stand in on a bare laptop). Still to decide: how the line becomes sound (retraced in the order and at the speed it was drawn, or read left to right) and what shaking it clear is.

After that:

- Machines: an EHCI driver, a text-mode UI for machines without a framebuffer, Realtek Ethernet (most desktops), and ports: Raspberry Pi, PowerPC Mac, coreboot payload.

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What the stick and the ISO carry besides BARE!

- **Limine**, the bootloader (its BIOS and UEFI files). Copyright (C) 2019-2026 Mintsuki and contributors. BSD 2-Clause License: [LICENSES/Limine-BSD-2-Clause.txt](#).
- **Terminus Font**, the text on screen (its 8x16 and 12x24 bitmaps, turned into C by `tools/psf2c.py`, with a few glyphs it lacks drawn in the same style). Copyright (C) 2020 Dimitar Toshkov Zhekov. SIL Open Font License 1.1: [LICENSES/Terminus-Font-OFL-1.1.txt](#).

Doom's engine

- **Doom** (id Software, 1993), whose source id released under the GPL in 1999, as **Chocolate Doom** (Simon Howard and contributors) and **doomgeneric** (ozkl) carry it: in `third_party/doom`, built into the kernels. Copyright (C) 1993-1996 id Software, Inc.; (C) 2005-2014 Simon Howard; (C) 1993-2008 Raven Software; and the others named in each file. GPL-2.0-or-later ([LICENSES/Doom-GPL-2.0.txt](#)); its `sha1.c` (from GnuPG, (C) 1998-2001 Free Software Foundation) is GPL-3.0-or-later. No game data comes with BARE!. DOOM is a trademark of id Software; BARE! is not made or endorsed by id Software.

Written from others' work

- **Ableton Link**: the protocol was implemented by reading Ableton's own source (github.com/Ableton/link, GPL-2.0-or-later) and checked against it. Ableton and Link are trademarks of Ableton AG; BARE! is not made or endorsed by Ableton.

Named after

Pages and sounds are named after the instruments and pieces that inspired them: Evgeny Murzin's ANS synthesizer, Iannis Xenakis's UPIC, GENDY3 and S.709, the Fairlight CMI, the Suzuki Omnichord, Ableton's Operator. The names describe; none of their makers is involved.



BARE! 2.5 beta · the manual, made from the README of github.com/willbearfruits/bare.
Printed or on screen, it is free like the instrument: GPL-3.0-or-later.